DECISION TABLE TESTING

A Case Study Across Common Business Applications

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Introduction

A decision point is the spot in an application where the system looks at one or more inputs and decides what should happen next. In decision table testing, these points are laid out systematically so that every realistic combination of conditions – and the action each one triggers – gets tested rather than assumed.

Key Points

- Decision points govern the path an application takes during execution.
- They are typically expressed as True/False, Yes/No, or Valid/Invalid checks.
- Each decision point can lead to one or more possible actions, depending on the combination of conditions.
- Mapping conditions and outcomes this way keeps complex business logic easy to follow.
- They are common across login screens, ATMs, e-commerce platforms, leave systems, and banking software.
- They help testers build thorough, systematic test cases that cover every relevant business rule.
- Working through decision points lowers the risk of overlooking an important scenario.
- They pin down exactly which action should follow which set of conditions.
- Both the positive and negative paths get verified before the software goes live.
- Overall, they raise the accuracy and dependability of the testing process.

Why Decision Points Are Useful

- They surface business rules that might otherwise be missed.
- They cut down on duplicate or overlapping test cases.
- They make the underlying decision logic easier to follow.
- They support a more structured, repeatable testing process.
- They widen overall test coverage.
- They contribute to a more dependable, consistent application.

Sample Decision Points

- Does the bank account hold enough balance?
- Has the manager signed off on the leave request?

- Has the order already been shipped?
- Did the payment go through successfully?
- Is the product currently in stock?
- Is the account still active?
- Is the password entered correct?
- Is the username entered valid?

1. Bank ATM Cash Withdrawal

CONDITIONS	RULES				
	R1	R2	R3	R4	R5
C1: Card Inserted	Y	Y	Y	Y	N
C2: Correct PIN Entered	Y	Y	N	N	Y
C3: Sufficient Balance	Y	N	Y	N	-
C4: Daily Limit Exceeded	N	N	N	Y	-
ACTIONS					
A1: Cash Dispensed	✓	✗	✗	✗	✗
A2: Show "Invalid PIN"	✗	✓	✗	✗	✗
A3: Show "Insufficient Balance"	✗	✗	✓	✗	✗
A4: Show "Daily Limit Exceeded"	✗	✗	✗	✓	✗
A5: Show "Insert Card"	✗	✗	✗	✗	✓

Note:
Y = Yes, N = No, - = Not applicable,
✓ = Action is executed,
✗ = Action is not executed

Conditions Considered

- C1: Card has been inserted
- C2: PIN entered is correct
- C3: Account has sufficient balance
- C4: Daily withdrawal limit already reached

Test Cases Derived from the Table

TC ID	Scenario / Input Combination	Expected Result	Rule Covered
TC3_01	Card inserted, correct PIN, balance available, limit not reached	Cash is dispensed	R1
TC3_02	Card inserted, wrong PIN entered	"Invalid PIN" message shown	R2
TC3_03	Card inserted, correct PIN, balance too low	"Insufficient balance" message shown	R3
TC3_04	Card inserted, correct PIN, daily limit already used up	"Daily limit exceeded" message shown	R4
TC3_05	No card inserted	"Insert card" prompt shown	R5

TC ID	Scenario / Input Combination	Expected Result	Rule Covered
TC3_06	Card inserted, wrong PIN, balance too low	“Invalid PIN” message shown	R6

Explanation of Each Test Case

TC3_01: Confirms cash is dispensed only when every condition is favourable.

TC3_02: Checks that an incorrect PIN stops the withdrawal before anything else is verified.

TC3_03: Confirms the machine checks the account balance before releasing cash.

TC3_04: Checks that withdrawals stop once the daily limit has already been used.

TC3_05: Confirms the ATM will not proceed with any transaction until a card is inserted.

TC3_06: Shows that PIN verification takes priority over the balance check when both fail.

2. E-Commerce Order Cancellation Policy

CONDITIONS	RULES				
	R1	R2	R3	R4	R5
C1: Order Placed	Y	Y	Y	Y	Y
C2: Order Shipped	N	Y	Y	Y	N
C3: Delivery Completed	N	N	Y	Y	Y
C4: Cancellation Within 24 Hrs	Y	N	-	-	-
C5: Return Policy Applicable	-	-	Y	Y	N
ACTIONS					
A1: Cancel Order	✓	✗	✗	✗	✗
A2: Cancel Not Allowed	✗	✓	✓	✗	✓
A3: Initiate Return / Refund	✗	✗	✓	✓	✗
A4: Show "Order Already Delivered"	✗	✗	✗	✓	✓
A5: Show "Return Not Applicable"	✗	✗	✗	✗	✓

Note:
Y = Yes, N = No, - = Not applicable,
✓ = Action is executed,
✗ = Action is not executed

Conditions Considered

- C1: Order has been placed
- C2: Order has already been shipped
- C3: Delivery is complete
- C4: Cancellation requested within 24 hours
- C5: Item qualifies under the return policy

Test Cases Derived from the Table

TC ID	Scenario / Input Combination	Expected Result	Rule Covered
TC5_01	Order placed, request made within 24 hours	Order cancelled successfully	R1
TC5_02	Order already shipped	Cancellation not permitted	R2
TC5_03	Order delivered, return window still open	Return process initiated	R3
TC5_04	Order delivered, return window has closed	Return request declined	R4
TC5_05	Order placed, request made after 24 hours	Cancellation declined	R5
TC5_06	Order shipped and delivered	Cancellation not permitted, return process applies instead	R6

Explanation of Each Test Case

TC5_01: Confirms an order can be cancelled while it is still within the free-cancellation window.

TC5_02: Confirms that once an order ships, cancellation is no longer an option.

TC5_03: Confirms customers can start a return while the return policy still applies.

TC5_04: Confirms a return is refused once the return window has lapsed.

TC5_05: Confirms cancellation requests made past the 24-hour mark are turned down.

TC5_06: Confirms that a delivered order can only be handled through the return flow, not cancellation.

3. Leave Approval Workflow

CONDITIONS	RULES					
	R1	R2	R3	R4	R5	R6
C1: Leave Days ≤ 2	Y	Y	N	N	-	-
C2: Leave Days > 2 & ≤ 5	N	N	Y	Y	-	-
C3: Leave Days > 5	N	N	N	N	Y	Y
C4: Manager Approved	Y	N	Y	N	Y	N
C5: HR Approved	-	-	Y	N	Y	N
ACTIONS						
A1: Leave Auto Approved	✓	✗	✗	✗	✗	✗
A2: Manager Approval Required	✗	✓	✓	✗	✗	✗
A3: HR Approval Required	✗	✗	✗	✓	✓	✓
A4: Leave Approved	✓	✗	✓	✗	✓	✗
A5: Leave Rejected	✗	✓	✗	✓	✗	✓

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Conditions Considered

- C1: Leave requested is 2 days or fewer
- C2: Manager has approved the request
- C3: HR has approved the request

Test Cases Derived from the Table

TC ID	Scenario / Input Combination	Expected Result	Rule Covered
TC4_01	Leave requested for 2 days	Leave gets auto-approved	R1
TC4_02	Leave for 4 days, manager approves	Leave approved	R2
TC4_03	Leave for 4 days, manager declines	Leave stays pending / rejected	R3
TC4_04	Leave for 7 days, manager and HR both approve	Leave approved	R4
TC4_05	Leave for 7 days, manager approves but HR declines	Leave rejected	R5
TC4_06	Leave for 7 days, manager declines	Leave rejected	R6

Explanation of Each Test Case

TC4_01: Confirms very short leave requests go through without needing manual sign-off.

TC4_02: Confirms a medium-length request goes through once the manager signs off.

TC4_03: Confirms a request stays blocked without the manager's approval.

TC4_04: Confirms longer leave needs sign-off from both the manager and HR.

TC4_05: Confirms HR's decision can override an approval already given by the manager.

TC4_06: Confirms a longer request is rejected outright if the manager does not approve it.

4. Online Shopping – Free Shipping Eligibility

CONDITIONS	RULES			
	R1	R2	R3	R4
C1: Order Amount ≥ ₹500	Y	Y	N	N
C2: User is Prime Member	Y	N	Y	N
C3: Product from Partner Seller	Y	N	Y	N
ACTIONS				
A1: Free Shipping	✓	✓	✗	✗
A2: Shipping Charges Applicable	✗	✓	✓	✓
A3: Show Eligible Message	✓	✗	✓	✗
A4: Show Not Eligible Message	✗	✓	✗	✓

Note:
Y = Yes, N = No, ✓ = Action is executed,
✗ = Action is not executed

Conditions Considered

- C1: Order value is ₹500 or more
- C2: Buyer holds a Prime membership
- C3: Item is sold by a partner seller

Test Cases Derived from the Table

TC ID	Scenario / Input Combination	Expected Result	Rule Covered
TC1_01	Order = ₹700, Prime member, partner seller	Free shipping applied	R1
TC1_02	Order = ₹700, Prime member, non-partner seller	Shipping charge applied	R2
TC1_03	Order = ₹700, not a Prime member, partner seller	Shipping charge applied	R3
TC1_04	Order = ₹400, Prime member, partner seller	Shipping charge applied	R4
TC1_05	Order = ₹500 exactly, Prime member, partner seller	Free shipping applied	R5
TC1_06	Order = ₹300, not a Prime member, non-partner seller	Shipping charge applied	R6

Explanation of Each Test Case

TC1_01: Covers the ideal case where every condition needed for free shipping is met.

TC1_02: Shows that items from a non-partner seller don't qualify, even for Prime members.

TC1_03: Shows that a large order alone isn't enough without an active Prime membership.

TC1_04: Confirms orders under the ₹500 mark are charged for shipping.

TC1_05: A boundary check confirming free shipping kicks in exactly at ₹500.

TC1_06: Confirms shipping is charged when none of the qualifying conditions apply.

5. User Login Validation

CONDITIONS	RULES			
	R1	R2	R3	R4
C1: Username Entered	Y	Y	Y	N
C2: Password Entered	Y	Y	N	Y
C3: Username Valid	Y	N	Y	N
C4: Password Valid	Y	N	Y	N
ACTIONS				
A1: Login Successful	✓	✗	✗	✗
A2: Show "Invalid Username"	✗	✓	✗	✗
A3: Show "Invalid Password"	✗	✗	✓	✗
A4: Show "Enter All Fields"	✗	✗	✗	✓

Note:
Y = Yes, N = No, ✓ = Action is executed,
✗ = Action is not executed

Conditions Considered

- C1: Username entered is valid
- C2: Password entered is valid
- C3: Account is currently active

Test Cases Derived from the Table

TC ID	Scenario / Input Combination	Expected Result	Rule Covered
TC2_01	Valid username, valid password, active account	Login succeeds	R1
TC2_02	Valid username, valid password, inactive account	"Account inactive" message shown	R2
TC2_03	Valid username, invalid password	"Invalid password" message shown	R3
TC2_04	Valid username, invalid password, inactive account	"Invalid password" message shown	R4
TC2_05	Invalid username	"Invalid username" message shown	R5
TC2_06	Invalid username, invalid password	"Invalid username" message shown	R6

Explanation of Each Test Case

TC2_01: Confirms login works when every credential checks out.

TC2_02: Confirms an inactive account is blocked even with the right credentials.

TC2_03: Confirms the system correctly flags an incorrect password.

TC2_04: Confirms password validation happens before the account-status check.

TC2_05: Confirms an unrecognised username is rejected right away.

TC2_06: Confirms username validation takes priority when both fields are wrong.